Historical Response Selection for Sampled FX Net-Position Peaks

A fixed comparison and an exploratory state-retrieval extension

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Abstract

Historical peak forecasts depend on which episodes are retained and how their responses are transferred to the current state. We compare empirical quantile constructions for the maximum absolute source-reported USD net position over the current and next H records, using an aggregate client-flow proxy from one foreign-exchange market maker. An original fixed comparison is extended by signed-state nearest-neighbor retrieval after the evaluation interval had been inspected. The primary comparisons use 6,625 common issue origins, a .95 quantile preference and fully completed labels; the extension selects 360 or 180 episodes from a rolling pool of 720. At H=6, historical headrooms have 2.08% lower mean pinball loss than raw current-anchor scenarios, with more breaches and larger overruns; the ranking reverses at H=24. In the extension's fixed primary comparison, nearest360 retrieval reduces floored native-peak-quantile loss by 31.83% relative to recent360 retrieval, but remains worse than the original full-window headroom and scenario rules. All 12 matched retrieval loss contrasts are negative, while every matched scenario contrast has more breaches and greater mean positive overrun. Elementary common-set bounds explain how nearby starting positions can make representations agree without certifying accuracy. The findings identify finite-record forecast tradeoffs, with exploratory retrieval evidence and no population superiority inference. The recorded proxy and passive forecasts do not establish conditional coverage, independent external validity or policy, profit and funding effects.

1. Introduction

Historical peak forecasts require two choices: which episodes to retain and how to transfer their responses to the current state. We compare native peaks (K), headrooms above each episode's starting magnitude (F), and signed increments reanchored at the current position (S). For a stipulated .95 quantile score and completed-label archive, we ask how replacing m recent episodes with m signed-position-nearest episodes changes finite-record loss and overrun/allowance tradeoffs for these three mature constructions.

The outcome is the largest absolute source-reported USD net position over the current and next H records, using sampled aggregate client flow from one foreign-exchange market maker. This proxy measures recorded net-position scale, rather than gross currency exposure, a hedged book, settlement cash or funding needs. The forecasts are passive: inventory-sensitive quotes affect future fills in dealer models (Avellaneda and Stoikov, 2008), but the policies behind these observations are unobserved. H counts subsequent source records, including those separated by 49-hour gaps, rather than a fixed elapsed-time risk window. Forecast scores do not estimate the effects of changing quotes, hedges or risk limits.

The original fixed comparison evaluates headrooms, reanchored increment envelopes and three simple references on 6,625 common issue origins. The extension was designed after those results had been inspected; its complete menu and primary contrast were recorded before the added forecasts were calculated. It crosses three representations with five historical sets. The primary contrast compares floored native-peak quantiles from the 360 nearest signed starting positions and the 360 most recent origins, selected from the same 720 fully completed episodes, at a six-record horizon. A second fixed count and a 24-record horizon provide sensitivity evidence. These remain exploratory comparisons on the inspected interval.

Common-set bounds connect differences between the three reports to historical starting-position distances. Exact signed-anchor matches make all three agree in real arithmetic. Nearest selection minimizes the largest selected signed distance among equal-count subsets; changes in actual report gaps and loss remain empirical questions. We report loss alongside issued levels, breaches, overrun, unused allowance, selected age and anchor distances. Construction agreement itself supplies no accuracy guarantee.

In the original primary comparison, F lowers loss by 2.08% relative to S, with more breaches and larger overruns; the ranking reverses at H=24. The extension lowers primary K loss by 31.83% relative to equal-count recent retrieval, a

weak baseline, while leaving the original full-window leaders ahead. Every matched S retrieval gain accompanies more breaches and greater mean positive overrun. The loss gains therefore have distinct breach and severity consequences across representations.

2. Closest work and mature method transfer

The closest empirical lineage is Wisniewski, Lindsay and Lindsay (2020), who already use nearest-neighbor regression on this source, alongside other regression engines, lagged positions and calendar or trading-session features, to forecast next-hour signed-position intervals. Their nearest-neighbor comparisons include quantile, conformal and normalized-conformal variants. Chapter 6 of Wisniewski's doctoral thesis continues same-source endpoint prediction with scale-normalized and regularized constructions (Wisniewski, 2024). Our multi-record absolute maximum includes the issue record, so their published interval scores are not numerical benchmarks for it. Their richer information sets also remain outside the present one-state comparison.

Financial path extrema are established forecasting and risk objects. MaxVaR studies the worst mark-to-market loss relative to initial wealth over a monitoring horizon (Boudoukh, Richardson, Stanton and Whitelaw, 2004). Maximum cumulative payment-cash deficits (van den Hoogen, 2017) and maximum collateral-adjusted contingent exposure with marking decisions (Liao and Theodosopoulos, 2005) address other peak-related financial questions. The present application differs in its sampled two-sided net-position response, state and units, rather than originating financial peak forecasting.

The transferred operations are also mature. Hull and White (1998) update historical shocks using predicted volatility; Barone-Adesi, Giannopoulos and Vosper (1999) propagate standardized innovations through conditional dynamics and portfolio revaluation. S instead translates raw observed net-position increments, without a fitted volatility filter or dynamics model. Biau and Patra (2011) construct nearest-neighbor empirical-response quantiles within a stationary-ergodic sequential strategy. We use the finite retrieval operation with a fixed window, one signed state and delayed, overlapping labels; its use does not transfer their consistency theorem or a coverage guarantee.

The contribution is a bounded application comparison: a multi-record peak response, completed-label timing, signed single-anchor retrieval crossed with response representation, and elementary common-set bounds explaining when those representations approach one another. It introduces neither nearest neighbors to this source nor a new estimator, independent dealer record or comparison against the strongest conditional models.

3. Data, recorded response and completed information

The input is the one-hour member of version 2 of the Net Position Time Series Dataset, with 18,576 records from 5 February 2014 01:00 through 24 January 2017 00:00. Wisniewski et al. (2020) describe sampled aggregate client-flow positions across five liquid FX pairs. Only datetime and NetPosUsd are used. A netted USD scalar does not reconstruct currency-level gross positions, hedge transactions, collateral or cash movements. The source's timezone and historical publication/vintage convention are unverified.

There are 18,420 one-hour and 155 forty-nine-hour adjacent gaps. We retain the file's chronology without sorting, deleting paths across gaps or interpolating missing intervals. Consequently H counts subsequent records, rather than elapsed hours. Let N_t be the signed source-reported position, observed before issue. Define

$$B_t=|N_t|;\quad P_{t,H}=\max_{0\le h\le H}|N_{t+h}|;\quad G_{t,H}=P_{t,H}-B_t.$$

Including h=0 gives the known floor B_t and nonnegative headroom G. The peak includes an interior recorded spike even if the final position returns to its starting value. It does not bound an unobserved excursion between records. The retrospective issue convention treats N_t as available immediately before issue, rather than establishing historical live-publication availability.

At issue t the extension's eligible pool is $I_t=\{t-H-719,...,t-H\}$. Each of its 720 historical origins i has a response ending at $i+H\le t$. The latest completed label may therefore include the already observed issue record. Earlier evaluation outcomes enter later pools only after completion. Adjacent peaks and rolling pools overlap; completed-label eligibility does not make them independent.

For each eligible origin retain its native peak $P_{i,H}$, headroom $G_{i,H}$ and directional increment extrema

$$a_{i,H}=\min_{0\le h\le H}(N_{i+h}-N_i);\quad b_{i,H}=\max_{0\le h\le H}(N_{i+h}-N_i);\quad a_{i,H}\le 0\le b_{i,H}.$$

These extrema determine the absolute peak after raw translation of the sampled increment path. They are not merely its two endpoints, and their deterministic sufficiency does not specify the next path's conditional law.

4. Historical selection and response representation

The extension uses five fixed sets at each issue. Recent sets take the last m eligible origins. Nearest sets take the m smallest signed-state distances $|N_i-N_t|$, preferring the most recent origin when distances tie. Historical response values and future query outcomes never resolve distance ties. Selected paired rows retain chronological order.

Table 1. Fixed historical sets drawn from the same 720 completed candidate episodes. The rank is k=ceil((m+1)19/20).

Set	Membership	m	k
full720	All eligible	720	685
recent360	Most recent	360	343
nearest360	Nearest signed state	360	343
recent180	Most recent	180	172
nearest180	Nearest signed state	180	172

For a selected set J of m rows, Q_k denotes the kth unweighted order statistic with duplicates retained. There is no interpolation or jitter; all configured corrected ranks are available. At these particular counts and $\tau=.95$ the selected order statistic is the upper endpoint of the empirical pinball-minimizer interval (Appendix C). This numerical rank convention establishes no coverage theorem for the dependent historical pool.

The three response representations issue the following levels, with H suppressed on historical labels:

$$F_J=B_t+Q_k\{P_i-|N_i|: i \text{ in } J\}.$$

$$S_J=B_t+Q_k\{\max(N_t+b_i,-N_t-a_i)-B_t: i \text{ in } J\}.$$

$$K_J=\max\{B_t,Q_k\{P_i: i \text{ in } J\}\}.$$

F retains headroom relative to historical starting magnitude; S translates signed increments to the current position; K retains native peaks subject to the current floor. K is the method labeled L in the original artifacts. The files call an issued forecast level C its 'capacity'; throughout the paper this is a passive reported level. All three already use current information: F/K use B_t, and S also uses signed N_t. Retrieval changes historical membership. Equal-count recent and nearest sets may differ in age, scale, regimes and dependence.

The original five-rule comparison also includes the radius reference $R=B_t+Q_k\{\max(b_i,-a_i)\}$ and the known-floor reference $P0=B_t$, on its full n-row pool. The extension implements only F/S/K. Stored float64 S subtracts the floor before ranking and adds it afterward, preserving the original operation order and its rounded decisions.

An elementary construction bound explains what matching can control. For one common nonempty set J and the same valid rank, put $r_{abs}=\max_{i \text{ in } J}||N_i|-B_t|$ and $r_{signed}=\max_{i \text{ in } J}|N_i-N_t|$. In real arithmetic,

$$|F_J-K_J|\le r_{abs};\quad |S_J-K_J|\le r_{signed};\quad |F_J-S_J|\le r_{abs}+r_{signed}\le 2r_{signed}.$$

Appendix A proves the bounds using floor projection and order-statistic stability. If every selected signed anchor equals the query, all three reports agree; if only every selected magnitude equals the query magnitude, F=K while S can differ. Plateaus can also give equal F/S reports at distant anchors. Nearest retrieval minimizes the largest selected signed distance among equally sized subsets, but need not minimize actual report differences or loss. These common-set bounds do not compare forecasts built from different selected histories, and they contain no statement about future calibration.

5. Design histories, score and reporting

The original specification was recorded at 05:46:37 UTC on 21 October 2026 and publicly committed before its target-specific forecasts and losses were calculated. It fixes H=6, n=720, τ=19/20 as primary, with six one-factor changes: H=12 or 24, n=480 or 1440, and τ=.90 or .99. The five rules and seven settings yield 35 original summary rows. This is a retrospective specification exercise with prior source-literature knowledge, not external preregistration or prospective collection.

The extension was conceived after those results had been inspected. Its human-readable protocol and hash-pinned machine specification were publicly recorded in commits 63703b9 and 69d5bb6 before the added forecasts. They fix H=6 and 24, the five sets in Table 1 and F/S/K: 15 configurations per horizon, 30 in total. The six full720 F/S/K rows repeat original F/S/L references. Recording the extension before execution limits further discretion but cannot turn the old interval into an unseen holdout.

Both designs use the same 6,625 issue origins: indices 11927 through 18551 inclusive, from naive source timestamp 1 January 2016 00:00 through 23 January 2017 00:00. The common original full-menu rule retains the future records required by H=24. Earlier evaluation labels become usable later only when completed. Changing H alters both the future peak and the end of the eligible historical archive, so cross-horizon comparisons do not isolate the maximum operator alone.

For stored error e=P−C, pinball loss is

$$\rho_\tau(e)=\tau \max(e,0)+(1-\tau)\max(-e,0); \quad \tau=19/20 \text{ in the extension.}$$

The underprediction weight is nineteen times the overprediction weight. These are stipulated forecast preferences, not estimated funding prices, shortage costs or regulatory coefficients. Quantile-consistent scoring targets the corresponding functional under suitable finite expected-loss conditions (Gneiting, 2011); it does not establish conditional correctness of an empirical historical rule. Original rows with different τ express different objectives and are not pooled into an across-preference winner.

The original primary contrast is mean loss(F) minus mean loss(S) in its primary setting. The extension's separate primary contrast is mean loss(nearest360 K) minus mean loss(recent360 K), at H=6; a nonnegative mean would fail its finite-record improvement criterion. H=24 repeats retrieval in a context adverse to the original F/S comparison, and m=180 is fixed sensitivity. Neither may replace an unfavorable primary contrast. All 12 nearest-minus-recent and 10 same-set F-minus-S contrasts are reported without a score-sign acceptance gate.

For an issued level C≥B, extra allowance is C−B. Positive overrun and unused allowance give the score decomposition

$$O=(P-C)^+; \quad U=(C-P)^+; \quad \rho_\tau(P-C)=\tau O+(1-\tau)U, \text{ where } z^+=\max(z,0).$$

Every configuration reports mean loss, level and extra allowance; strict breach count/rate; mean O; conditional overrun with its own breach denominator; and mean U. Strict breach is P>C, with no tolerance changing its sign. Conditional overrun averages O only over that rule's breach events; zero breaches make it undefined. The target can have an atom at B, so even a true quantile's strict exceedance need not equal 1−τ. Nominal breach proximity is not a calibration certificate.

Geometry reports both anchor radii, mean selected age in records, the mean absolute F/S gap and exact equal-capacity counts. Primary monthly retrieval contrasts retain all 13 inclusive calendar months at both H; groups below 30 origins retain counts/status while suppressing metrics. These are descriptive groups, not independent replications or an adaptive strategy. No new population test, confidence interval or bootstrap is attached to the extension. Original block-resampling profiles concern only the original full720 F/S contrast; Appendix B preserves their descriptive status.

6. Original comparison: score advantage with adverse tradeoffs

Table 2. Original primary H=6, n=720, τ=.95 results. Monetary columns are source-reported USD millions; breach is a percentage. K is the original L. Every rule has 6,625 origins.

Rule	Loss	Level	Breaches	Breach %	Overrun	Unused
F	0.441774	22.645674	332	5.011	0.166531	5.671391
S	0.451149	23.829232	254	3.834	0.116728	6.805145
R	0.527544	26.400452	122	1.842	0.064562	9.324199
K (old L)	0.925885	31.731323	489	7.381	0.196359	14.786868
P0	1.813620	15.231741	5102	77.011	1.909073	0.000000

Historical headrooms F have mean loss USD 441,774.04 versus USD 451,148.90 for raw reanchored scenarios S. The paired mean is −USD 9,374.86, or −2.08% of S loss. It is aggregated from per-origin loss differences before display rounding. All original sensitivities and methods are retained in Appendix B and the public aggregates.

The lower score accompanies 332 strict breaches for F versus 254 for S, and greater mean positive overrun (USD 166,531.06 versus USD 116,728.04). Conditional overrun is USD 3.323097 million over 332 events for F and USD 3.044580 million over 254 for S. Lower unused allowance offsets the larger overrun at the stipulated weights. It is a score tradeoff rather than safety dominance or a measured financial saving.

R has fewer breaches and smaller unconditional overrun, but larger mean capacity and score. K has the greatest mean capacity yet 489 breaches, illustrating why an average capacity ordering does not imply pathwise dominance. P0 has zero extra and unused allowance, but 5,102 breaches and the largest loss. Conditional means must be read with these different event sets, rather than ranked as if they describe the same tail observations.

The F/S direction reverses at H=24: F loss is USD 777,068.84 and S loss USD 748,383.54, with paired mean +USD 28,685.30. The τ=.90 difference is also positive but small (+USD 144.23), retained without rounding it to zero or declaring equivalence. Original monthly means favor F eight times and S five times. Low and middle pre-cutoff magnitude groups favor S, while the high group favors F and accounts for 5,445 of 6,625 origins. Those descriptive composition patterns do not identify a causal state effect or constitute a formal stationarity test.

All four original geometric-block resampled-mean ranges cross zero. They are descriptive profiles, not population confidence intervals, superiority probabilities or proof of equality. The original H24 reversal and these limits remain unchanged by a later favorable retrieval contrast, which asks a different question.

7. Exploratory historical selection

The primary paired mean is negative on this inspected interval. At H=6, m=360, nearest360 K loss is USD 594,811.09 versus USD 872,515.55 for recent360 K; the direct paired difference is −USD 277,704.47, a 31.83% reduction relative to recent360 K. Both original full720 F and S remain lower. At H=24, nearest360 K improves over recent360 K from USD 1,246,630.42 to USD 960,696.45, while original full720 S remains lower at USD 748,383.54. The large relative gain therefore improves a weak native-level baseline, without establishing a new best forecast.

Table 3. Complete extension loss menu and same-set F-minus-S paired mean, in source-reported USD millions. Full720 rows repeat original references. Differences are computed before display rounding.

H	Historical set	F loss	S loss	K loss	F−S mean
6	full720	0.441774	0.451149	0.925885	-0.009375
6	recent360	0.454089	0.467678	0.872516	-0.013589
6	nearest360	0.448664	0.448731	0.594811	-0.000067
6	recent180	0.485866	0.478940	0.785878	+0.006926
6	nearest180	0.468133	0.459592	0.525471	+0.008541
24	full720	0.777069	0.748384	1.197833	+0.028685
24	recent360	0.807496	0.824047	1.246630	-0.016551
24	nearest360	0.791192	0.770354	0.960696	+0.020838
24	recent180	0.858214	0.895718	1.284092	-0.037504
24	nearest180	0.847734	0.827305	0.963931	+0.020429

All 12 fixed nearest-minus-recent score contrasts are negative (Figure 2). Absolute reductions are largest for K in each horizon/count pair: about USD 260,407–320,161, compared with USD 18,947–68,413 for S and USD 5,425–17,733 for F. These are correlated comparisons sharing outcomes and historical pools, not 12 independent confirmations. Nearest360 S at H=6 slightly improves its full720 S counterpart. Nevertheless, no added configuration beats the better original full720 F/S rule at the corresponding H: F at H=6 and S at H=24. This observed menu comparison is not a new prespecified model-selection criterion.

For the primary K pair, mean capacity falls from USD 29.854484 million to USD 25.224925 million; mean unused allowance falls from USD 12.950501 million to USD 8.274716 million. Breaches fall from 637 to 507 (9.615% to 7.653%), and mean positive overrun falls from USD 236,832.10 to USD 190,605.57. Conditional overrun nevertheless rises from USD 2.463128 million over 637 events to USD 2.490655 million over 507. The denominators and breach sets differ. The improved empirical breach rate remains above the nominal 5% reference; this descriptive comparison is not a population calibration test. None of these quantities is an observed funding payment or conditional-calibration result.

Every matched S retrieval comparison has the opposite breach/overrun tradeoff: lower loss but more strict breaches and greater mean positive overrun. At H=24, m=360, nearest S loss falls from USD 824,046.89 to USD 770,353.74, while breaches rise from 320 to 430 and mean positive overrun from USD 190,842.79 to USD 283,299.56. Lower capacity and unused allowance offset the overrun increase under the score. The same adverse direction holds for both counts at both horizons. Describing all 12 score gains as uniform risk reduction would be inaccurate.

For F, nearest retrieval reduces breach counts and mean positive overrun in all four matched comparisons, but conditional mean overrun rises in all four. At H=24, m=180, it rises from USD 4.650289 million over 588 breaches to USD 5.349645 million over 454. These conditional means summarize different breach sets; they do not identify an increase in severity on the same events.

At a fixed count, nearest membership guarantees a weakly smaller signed radius in the finite selection problem. Report gaps, exact equalities and score rankings remain empirical outcomes. At H=6, m=360, mean signed radius falls from USD 30.399817 million for recent retrieval to USD 12.365836 million for nearest retrieval, mean absolute F/S gap from USD 1.256683 million to USD 0.477250 million, and exact equal-capacity count rises from 1,804 to 4,297. The paired F−S loss mean becomes −USD 66.78, rather than recent360's −USD 13,589.40. Its small magnitude is an observed difference, not statistical equivalence.

At H=24, m=360, the smaller nearest F/S report gap coexists with a loss-order reversal: recent360 favors F by USD 16,550.54, whereas nearest360 favors S by USD 20,837.78. Reducing nearest count from 360 to 180 makes radii and F/S gaps smaller at both H but makes both F and S losses larger. Thus geometric agreement cannot select a predictive optimum. The complete geometry and age table is retained in Appendix B.

Nearest rows are older on average than equal-count recent rows. At m=360, mean age is 328.35 versus 185.50 records for H=6, and 352.09 versus 203.50 for H=24; at m=180 the nearest ages are 301.27 and 331.82 versus 95.50 and 113.50. State proximity, age and potentially regime/scale/dependence composition change together. The contrast compares two complete membership rules; it does not separate proximity from the accompanying age and composition changes.

The primary monthly K contrast favors nearest retrieval in 12 of 13 months at each H (Figure 3). January 2016 is adverse at both: +USD 3,379.20 for H=6 and +USD 520,447.46 for H=24. All month counts exceed the fixed display threshold. Origin-count-weighted monthly means reconcile to the global paired means; months remain descriptive groups within the same record, and the adverse month is not excluded.

8. Interpretation and limits

Nearest-state retrieval lowers loss relative to equal-count recent retrieval throughout the fixed menu, with the largest absolute gains for K. The tradeoffs differ by representation. All four matched F comparisons have fewer breaches and lower mean positive overrun, but higher conditional mean overrun. The primary K comparison also has fewer breaches and lower mean positive overrun, but higher conditional mean overrun. Every matched S comparison has more breaches and greater mean positive overrun. Conditional means use each method's own breach set. The original full-window leaders remain ahead, and the one-state rule is not benchmarked against the richer information in Wisniewski et al. (2020), Wisniewski (2024), conditional regressions or fitted filtered historical simulation.

Smaller representation gaps accord with the finite geometry, without identifying an accurate conditional forecast. Raw translation also differs from conditional path generation. Appendix A's known-linear-dynamics illustration gives a horizon-dependent anchor correction for innovation replay, whereas raw translation retains the full anchor difference at every step. This deterministic illustration fits no coefficient and establishes no inventory-control mechanism for the source.

The declared 19:1 loss ratio allows less unused allowance to offset greater overrun. Another loss or origin weighting asks a different decision question; funding prices and shortage costs are unobserved. Even accurate prediction of the sampled net-position proxy would not establish cash adequacy, gross exposure, P&L or a changed policy's causal effect.

The evidence comes from one aggregate 2014–2017 record. Reexecution, an identical-hash download and a same-source thesis support reproducibility, not transfer across dealers or regimes. Independent inventory data require an audit of aggregation, units, sampling and release timing; price returns and weekly position reports measure different objects. Overlapping maxima, shared episodes and persistent positions also create dependence. Stationarity, mixing, conditional-law and tail-information conditions for broader inference remain unestablished. Fixed neighbor counts do not count independent tail observations, and geometry, empirical ranks and breach frequencies do not establish coverage. The extension remains exploratory because its interval had already been inspected.

9. Conclusion

Historical selection and response representation produce distinct forecast tradeoffs for sampled client-flow net-position peaks. Original headroom pooling lowers primary loss by 2.08% relative to raw scenario reanchoring, with more breaches and greater overrun; the ranking reverses at H=24. Exploratory state retrieval improves equal-count recent rules, including a 31.83% primary reduction for the weak native-peak baseline, while the original full-window leaders remain ahead. Every matched scenario gain entails more breaches and greater mean positive overrun. Geometry explains construction agreement without deciding predictive accuracy. Independent dealer records and same-target conditional benchmarks are needed to assess transfer beyond this application.

Appendix A. Finite geometry and its predictive limits

A.1. Common-set projection bounds

These are elementary finite-array arguments, rather than new asymptotic or coverage results. Fix a nonempty common selected set J, an available rank $1\le k\le|J|$, query $x=N_t$, floor $B=|x|$ and completed historical windows. All identities in this appendix concern real arithmetic. Write $B_i=|N_i|$, $d_{i,h}=N_{i+h}-N_i$, with $d_{i,0}=0$, and define

$$A_i=B+P_i-B_i;\quad T_i=\max_{0\le h\le H}|x+d_{i,h}|;\quad C_B(z)=\max(B,z).$$

Both A_i and T_i are at least B. The projection C_B is nondecreasing and 1-Lipschitz. It commutes with a finite order statistic, as does a common translation. Hence $F_J=Q_k(A)$, $S_J=Q_k(T)$ and $K_J=Q_k(C_B(P))$. Since $C_B(A_i)=A_i$,

$$|A_i-C_B(P_i)|\le|A_i-P_i|=|B-B_i|.$$

The reverse triangle inequality bounds each path value's change under a starting-position translation by $|x-N_i|$. Taking a maximum gives $|T_i-P_i|\le|x-N_i|$. Applying the same floor projection preserves that bound because $T_i\ge B$. Finally, paired order statistics are nonexpansive in the maximum coordinate difference: if $|u_i-v_i|\le r$ for every i, then $u_i\le v_i+r$ and conversely; monotonicity and common translation give $|Q_k(u)-Q_k(v)|\le r$. These observations prove the two K bounds in Section 4. The triangle inequality and $||N_i|-|x||\le|N_i-x|$ prove the stated F/S bound. Keeping the rank fixed while changing one side's historical rows does not by itself justify these common-set comparison bounds.

Among all size-m subsets, selecting the m smallest signed distances minimizes the largest selected distance: any alternative must include an element at least as far away as the mth ordered distance. This finite minimax fact, with the declared tie convention, concerns the signed radius. It does not minimize the absolute-anchor radius, the sum of the two radii, the actual report gap or a forecast loss. For every fixed realized y, pinball loss is $\max(\tau,1-\tau)$-Lipschitz in its report, so a report-gap bound also bounds the absolute loss difference. Such stability supplies no sign for the realized or expected loss difference.

A.2. Envelope plateaus and the role of signed anchors

Put $u_i=b_i\ge0$ and $v_i=-a_i\ge0$. The headroom of an increment envelope at any anchor z is

$$g_i(z)=\max(z+u_i,-z+v_i)-|z|;\text{ for } z\ge0 \text{ it equals } \max(u_i,v_i-2z), \text{ and for } z\le0 \text{ it equals } \max(u_i+2z,v_i).$$

The branches agree at zero. Each is 2-Lipschitz on its half-line, so the whole function is 2-Lipschitz; its range lies in $[\min(u_i,v_i),\max(u_i,v_i)]$. Applying these two bounds to $g_i(N_i)$ and $g_i(x)$, and then order-statistic stability, gives

$$|F_J-S_J|\le\max_{i\text{ in }J}\min\{2|x-N_i|,|u_i-v_i|\}.$$

For example, $u_i=v_i$ in every row makes $F_J=S_J$ at arbitrary anchors. Nonnegative query/source anchors together with $u_i\ge v_i$ in every row also suffice; the nonpositive analogue uses $v_i\ge u_i$. These are sufficient conditions, not a characterization: pooled order statistics can coincide even when paired rows differ. The range bound also gives $F_J,S_J\le R_J$, where $R_J=B+Q_k\{\max(u_i,v_i):i\text{ in }J\}$. This is a theoretical same-set ordering; no R retrieval configuration is evaluated in the extension, and the ordering does not make R_J preferable under a given loss.

If every $B_i\le B$, then $A_i\ge C_B(P_i)$ and $F_J\ge K_J$. If every $B_i\ge B$, then $C_B(P_i)=P_i\ge A_i$ and $F_J\le K_J$. Equal absolute anchors force F=K but leave signs relevant to S. A coherent full-pool example uses c>0, H=1, n=720, $\tau=19/20$, t=721 and $N_s=(-1)^{s+1}c$ for $0\le s\le721$. The mature pool is $J=\{1,...,720\}$, with 360 anchors of each sign and current $N_t=c$. All native peaks equal c and headrooms equal zero, so F=K=c. Raw reanchoring gives 360 peaks c and 360 peaks 3c; rank 685 gives S=3c. Thus $r_{abs}=0$ while $r_{signed}=2c$ and |S−K|=2c. This artificial path respects maturity and an available corrected rank; it is not an empirical claim or a claim about which rows nearest retrieval selects.

A.3. The same observed history allows opposite loss rankings

Choose d>0, H≥1, $0<\tau<1$ and n large enough that $k=\text{ceil}((n+1)\tau)\le n$. Set t=H+n−1 and $N_s=d(t-s)$ for $0\le s\le t$, using $J=\{0,...,n-1\}$. Every selected response ends by t. Each historical peak is its own positive starting value, so $G_i=0$ and F=0. Each increment path equals −dh; reanchoring at $N_t=0$ gives S=Hd. Two continuations of this identical observed history yield:

Table A1. Artificial coherent continuations. Symbols are constructed quantities, not source-record estimates.

Future N(t+h), 1≤h≤H	Actual peak	F loss	S loss
0	0	0	(1−τ)Hd
−dh	Hd	τHd	0

Either report can therefore have strictly smaller realized loss under a compatible future. Even a capacity ordering need not hold universally: the artificial history (0,−4,−1,2), with H=1, n=3, $\tau=3/4$ and issue t=3, has rank 3, F=6 and S=5.

These constructions establish logical limits, not the conditional law of the recorded series or a population ranking. A deterministic construction can refute an unconditional mathematical assertion while supplying no empirical validation for either rule.

A.4. Raw translation versus replay under known dynamics

For interpretation only, suppose $N_{s+1}=\mu+\varphi(N_s-\mu)+\varepsilon_{s+1}$, with a common known center and coefficient. Replay a historical window's ordered innovations from query x. Subtracting the two recursions and using induction gives

$$M_{i,h}(x)=N_{i+h}+\varphi^h(x-N_i); \quad W_{i,h}(x)=N_{i+h}+(x-N_i).$$

Taking the maximum absolute path value yields

$$|\max_h|W_{i,h}(x)|-\max_h|M_{i,h}(x)||\le\max_{0\le h\le H}|1-\varphi^h|\,|x-N_i|.$$

For $0\le\varphi\le1$ the coefficient is $1-\varphi^H$; at $\varphi=1$ the paths coincide. The identity needs no innovation law, while treating past innovations as representative of future conditional innovations is a separate assumption. Neither the recursion nor that assumption is fitted or validated here. Except in special cases such as $\varphi=1$, evaluating the horizon-dependent correction generally requires ordered path information that increment extrema discard. Thus the extrema are sufficient for exact raw translation, not for arbitrary filtered simulation or correct conditional path generation.

Appendix B. Complete settings and descriptive diagnostics

The following tables retain favorable, adverse and small differences. Monetary entries are source-reported USD millions, rounded only for display. Paired means come directly from their accepted aggregate columns; the tables do not recompute forecasts, select settings or create new tests. K denotes original L. Original settings change one factor from H=6, n=720, τ=.95: h12/h24 change H, n480/n1440 change n, and tau90/tau99 change τ. All use the same 6,625 issue origins. Different τ values express different score preferences.

Table B1. Complete original score grid. Δ is the stored paired mean F minus S; negative favors F within that setting.

Setting	F	S	R	K	P0	Δ F−S
main	0.441774	0.451149	0.527544	0.925885	1.813620	-0.009375
h12	0.592518	0.604781	0.689604	1.058614	2.950594	-0.012263
h24	0.777069	0.748384	0.803106	1.197833	4.630827	+0.028685
n480	0.449843	0.460816	0.532515	0.885243	1.813620	-0.010973
n1440	0.443793	0.448275	0.526979	0.931562	1.813620	-0.004482
tau90	0.659643	0.659499	0.763481	1.477201	1.718166	+0.000144
tau99	0.154991	0.158998	0.175486	0.288834	1.889983	-0.004007

The original groups were fixed using issue-time information except the explicitly retrospective future-gap panel. Magnitude cutoffs use absolute positions strictly before the evaluation interval; neither cutoff is reestimated within an evaluation group. Sign and magnitude panels have unequal membership. The index-modulo-six panel is a mechanical phase diagnostic rather than a six-hour sampling experiment. A group below 30 origins retains its count and suppression status; an empty current-zero group does not have a zero mean loss.

Table B2. All original noncalendar paired diagnostic groups. Δ is F minus S at the original primary setting. 'Not displayed' denotes insufficient membership, not an estimated zero.

Panel	Group	Origins	Δ F−S
Sign	negative	5592	-0.010808
Sign	zero	0	Not displayed
Sign	positive	1033	-0.001618
Magnitude	low	420	+0.023570
Magnitude	middle	760	+0.016090
Magnitude	high	5445	-0.015470
Index mod 6	0	1104	-0.012793
Index mod 6	1	1104	-0.016473
Index mod 6	2	1104	-0.005393
Index mod 6	3	1104	-0.005242
Index mod 6	4	1104	-0.005957
Index mod 6	5	1105	-0.010390
Future gaps	gap_free	6289	-0.006642
Future gaps	has_gap	336	-0.060535

Gap-crossing groups classify the realized future path and cannot be used as an issue-time filter. The 336 gap-crossing origins contain only one F breach and one S breach; their conditional severities rest on one event each. Calendar panels use naive source timestamps. Table B3 retains the original F/S month contrast and breach counts alongside the extension's two primary K month contrasts; they are separate contrasts, not interchangeable strategies.

Table B3. All months. Old Δ is original H6 F minus S; new Δ is nearest360 K minus recent360 K. Breach counts belong to old F and S. Every month meets the display threshold.

Month	Origins	Old Δ	F breaches	S breaches	New Δ H6	New Δ H24
2016-01	504	-0.017183	29	18	+0.003379	+0.520447
2016-02	504	-0.015726	21	18	-0.366978	-0.406803
2016-03	552	+0.040380	42	31	-0.211499	-0.086232
2016-04	502	+0.020270	37	24	-0.522837	-0.589971
2016-05	530	-0.031114	10	7	-0.432595	-0.352734
2016-06	528	-0.035273	16	7	-0.236362	-0.105785
2016-07	504	+0.006976	34	27	-0.213175	-0.494943
2016-08	552	-0.047937	14	12	-0.494498	-0.685697
2016-09	526	+0.002870	31	29	-0.236850	-0.338212
2016-10	506	-0.007725	39	36	-0.230552	-0.135516
2016-11	528	+0.007843	25	24	-0.220045	-0.310768
2016-12	526	-0.032346	15	11	-0.323571	-0.422938
2017-01	363	-0.012823	19	10	-0.041940	-0.286619

Table B4. All extension geometry rows. Mean per-origin anchor radii and mean absolute report gaps are in USD millions; age is mean t−i in records, not hours; equality counts are exact stored F=S comparisons.

H	Set	Abs radius	Signed radius	|F−S|	Equal	Age
6	full720	28.439284	36.604661	1.212343	809	365.50
6	recent360	24.262721	30.399817	1.256683	1804	185.50
6	nearest360	11.346111	12.365836	0.477250	4297	328.35
6	recent180	20.063962	24.372598	1.249574	2914	95.50
6	nearest180	6.506279	7.065549	0.313294	5100	301.27
24	full720	28.603627	36.864344	1.836659	1234	383.50
24	recent360	24.420571	30.694525	2.824054	2412	203.50
24	nearest360	11.566355	12.627357	0.703643	4350	352.09
24	recent180	20.576841	25.146348	3.169128	3342	113.50
24	nearest180	6.782036	7.387771	0.680282	5062	331.82

All ten geometry rows pass the per-origin declared tolerance checks. Nine aggregate maximum excesses are zero; H24/nearest180 records a maximum excess of 1.862645149230957e−9 USD, reflecting floating-point arithmetic rather than an exact-real violation. Its maximum recorded arithmetic tolerance is 1.041112061415106e−6 USD. Comparing these two aggregate maxima alone is not the per-origin verification; the issuer checks every origin and the accepted run records that check. Tolerances do not change breach flags, exact-capacity equality or loss signs.

The original descriptive resampling operates on the single already realized primary F-minus-S loss-difference sequence. For each length ℓ in {120,240,720,1440}, it uses 999 circular resamples: start uniformly at an origin, draw a positive geometric block length with mean ℓ, continue circularly, and truncate only the final block to length M=6,625. Each length uses NumPy PCG64 with SeedSequence([20261020,ℓ]). The table reports ddof=1 resampled standard deviation and noninterpolated order statistics 25 and 975. Constant input is preserved exactly; no recentering or model refitting is performed.

Table B5. All four original resampled-mean profiles, in USD millions where monetary. Expected block counts include the final truncated block. The displayed ranges are not population confidence intervals.

ℓ	ℓ/M	E blocks	Rep. mean	Rep. SD	Order 25	Order 975
120	0.0181	56.2	-0.009774	0.009310	-0.028186	+0.008559
240	0.0362	28.6	-0.009835	0.008948	-0.028279	+0.008127
720	0.1087	10.2	-0.009471	0.007557	-0.025106	+0.004780
1440	0.2174	5.6	-0.009475	0.006679	-0.023037	+0.003596

Politis and Romano (1994) supply the stationary-bootstrap lineage, whose formal results impose conditions on the stochastic process and block-length asymptotics. Here no relevant stationarity/mixing conditions have been established, and large fixed ℓ/M leaves few blocks. The profiles therefore describe a fixed resampling mechanism's sensitivity on this record. They are not applied to any added nearest-state contrast, and their endpoints are neither null-test cutoffs nor superiority probabilities.

Appendix C. Rank, arithmetic and replication conventions

For m sorted responses $z_{(1)}\le\ldots\le z_{(m)}$, the corrected rank is k=ceil((m+1)τ). If k>m, this implementation stops rather than inserting an infinite report or changing the rank. All declared cells have an available rank. The original seven ranks are 685,685,685,457,1369,649,714 in the Table B1 order. In the extension τ=19/20 and m in {720,360,180}, so mτ in {684,342,171} is an integer. The empirical pinball-minimizer interval is $[z_{(m\tau)},z_{(m\tau+1)}]$, including ties; the configured corrected rank chooses its upper endpoint. This particular relationship is not asserted for the original τ=.99 cell, nor does any empirical-minimizer identity establish future coverage.

All forecasts use stored float64 values, duplicate-preserving order statistics and the original operation order. For S, the translated peak is computed first, B is subtracted before ranking, and B is added afterward. Replacing this with a mathematically equivalent differently rounded expression can alter stored capacity ties, strict flags and scores. The scientific tables use direct error P−C; the paired summary uses compensated aggregation of per-origin loss differences, rather than subtracting displayed or independently rounded global means. Empty CSV values and JSON nulls have documented undefined reasons. Finite numerical reconciliation tolerances are kept separate from event definitions.

The extension's per-origin geometry tolerance is 64ε times the maximum of one, current floor, historical anchor magnitudes, native peaks, all three reports and the two anchor radii. The input source, specification and code identities are checked before accepting a run and rechecked after computation. Membership is computed from signed source positions at the completed historical origins; absolute magnitude alone cannot authenticate their signs. Selected age and radii are descriptive properties of that membership, not extra tuned predictors.

The input is version 2's one-hour member, 1hourNetPositionsKaggle.csv, with 18,576 records and 1,286,391 bytes. Its accepted SHA-256 is b8e92c54a76b49990b028252d6a89195e92193990e72f10228f2280a11a967fa. The public Net Position Time Series Dataset is identified by the Kaggle owner/slug davidlindsay1979/net-position-time-series-dataset. The source fingerprint, member/version, acquisition checks and result/code manifests allow identity verification. Public availability is not taken as permission to redistribute the raw observations; this repository supplies acquisition instructions and analysis code.

The original accepted computation took 8.02145 seconds, with peak sampled process RSS 80,936,960 bytes and one observed OS thread. The extension's sole accepted calculation took 10.307238 seconds on its internal process clock, with peak sampled RSS 64,602,112 bytes and one observed thread over 675 snapshots. A process-completion measurement reported 10.35 seconds and maximum RSS 64,634,880 bytes; its scope and precision differ from the internal measurement. These are execution observations under the single-process CPU budget, not general runtime benchmarks. The extension reused the exact accepted local input and did not download or redistribute observations.

The original results remain in results/summary.csv, paired-comparison.csv, stability.csv and diagnostics.csv with their original manifest. The separately frozen extension has summary.csv (30 rows), geometry.csv (10), comparisons.csv (22), monthly.csv (26) and manifest.json under results/revision-20261021/. Six shared reference rows reproduce the original F/S/L numeric columns exactly, with K mapped to L. Reproducing these values on the same hash-identified source verifies execution, not external validation. The root README documents separate acquisition, scientific execution, aggregate-only figure construction and manuscript rebuilding. Document rendering reads approved text and figure assets rather than running financial experiments.

References

Avellaneda, M. and S. Stoikov (2008). High-frequency trading in a limit order book. *Quantitative Finance* 8(3), 217–224. DOI:10.1080/14697680701381228.

Barone-Adesi, G., K. Giannopoulos and L. Vosper (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets* 19(5), 583–602. DOI:10.1002/(SICI)1096-9934(199908)19:5<583::AID-FUT5>3.0.CO;2-S. Consulted body: USI Quaderno 99-04, revised March 1999; final-body equivalence is unverified.

Biau, G. and B. Patra (2011). Sequential Quantile Prediction of Time Series. *IEEE Transactions on Information Theory* 57(3), 1664–1674. DOI:10.1109/TIT.2011.2104610. Consulted body: arXiv:0908.2503v2, 31 May 2010; final-body equivalence is unverified.

Boudoukh, J., M. Richardson, R. Stanton and R.F. Whitelaw (2004). MaxVaR: Long-Horizon Value at Risk in a Mark-to-Market Environment. *Journal of Investment Management* 2(3), 14–19. Consulted author manuscript; its cover is undated and final-body equivalence is unverified.

Gneiting, T. (2011). Making and Evaluating Point Forecasts. *Journal of the American Statistical Association* 106(494), 746–762. DOI:10.1198/jasa.2011.r10138.

van den Hoogen, S. (2017). Intraday liquidity risk estimation using transaction data: an extreme value theory approach. MSc Applied Mathematics thesis, Delft University of Technology, August 2017. Institutional source.

Hull, J. and A. White (1998). Incorporating volatility updating into the historical simulation method for value-at-risk. *Journal of Risk* 1(1), 5–19. DOI:10.21314/jor.1998.001. Consulted body: the author's October 1998 manuscript; final-body equivalence is unverified.

Liao, J. and T.V. Theodosopoulos (2005). Study on optimal timing of mark-to-market for contingent credit risk control. arXiv:math/0506077v1, 4 June 2005. The PDF names Theodore V. Theodosopoulos; arXiv metadata uses Ted Theodosopoulos.

Politis, D.N. and J.P. Romano (1994). The Stationary Bootstrap. *Journal of the American Statistical Association* 89(428), 1303–1313. DOI:10.1080/01621459.1994.10476870.

Wisniewski, W., D. Lindsay and S. Lindsay (2020). Application of Conformal Prediction Interval Estimations to Market Makers' Net Positions. Proceedings of the Ninth Symposium on Conformal and Probabilistic Prediction and Applications, *Proceedings of Machine Learning Research* 128, 285–301. Official proceedings.

Wisniewski, W.J. (2024). Net Position and Retail Trader Behaviour: Implications for Market Makers' Risk Management Strategies. Doctoral thesis, Royal Holloway, University of London. Institutional original. The institutional award year is 2024; the inspected cover is dated 2023.

davidlindsay1979. *Net Position Time Series Dataset*, version 2. Kaggle. Dataset record. Source identity verified 21 October 2026; raw redistribution permission remains unverified.

Figure captions

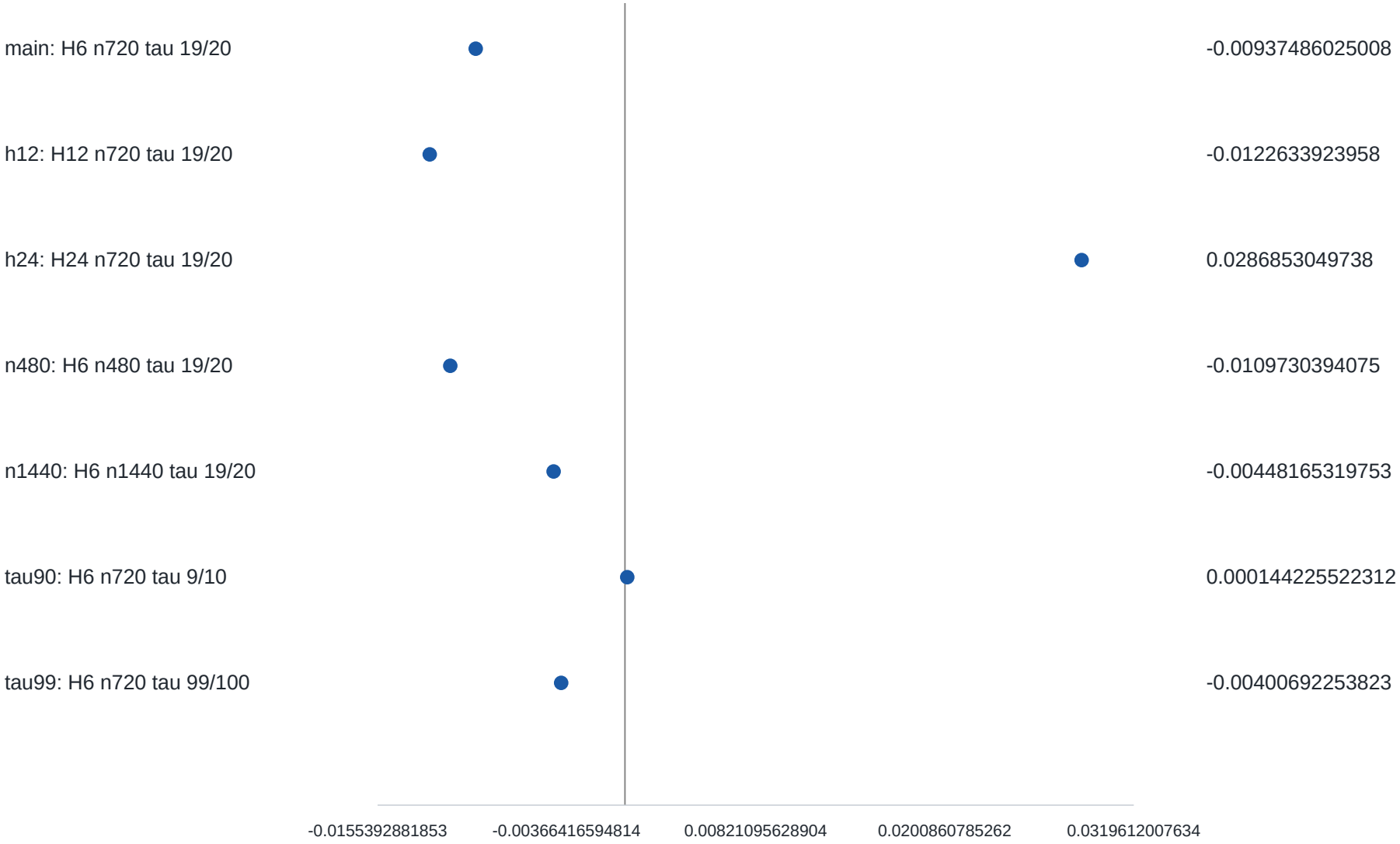
Figure 1. Original seven-cell F-minus-S paired means. Negative values favor F for that fixed preference. Different quantile preferences are not pooled into an overall winner.

Figure 2. All 12 fixed exploratory nearest-minus-recent loss contrasts. Negative values favor nearest retrieval at the same horizon, count and representation. The marked primary is H6, m360, K; none of these contrasts establishes superiority over the original full-window leaders.

Figure 3. Every monthly primary K retrieval contrast at both horizons, including adverse January 2016. Counts are source-record origins. Calendar groups are descriptive, not independent validations or switching regimes.

Within-cell paired mean loss differences

F minus S; USD millions. Negative values favor F within the same fixed cell.



Seven declared cells in fixed order; no comparison of raw losses across different tau objectives.

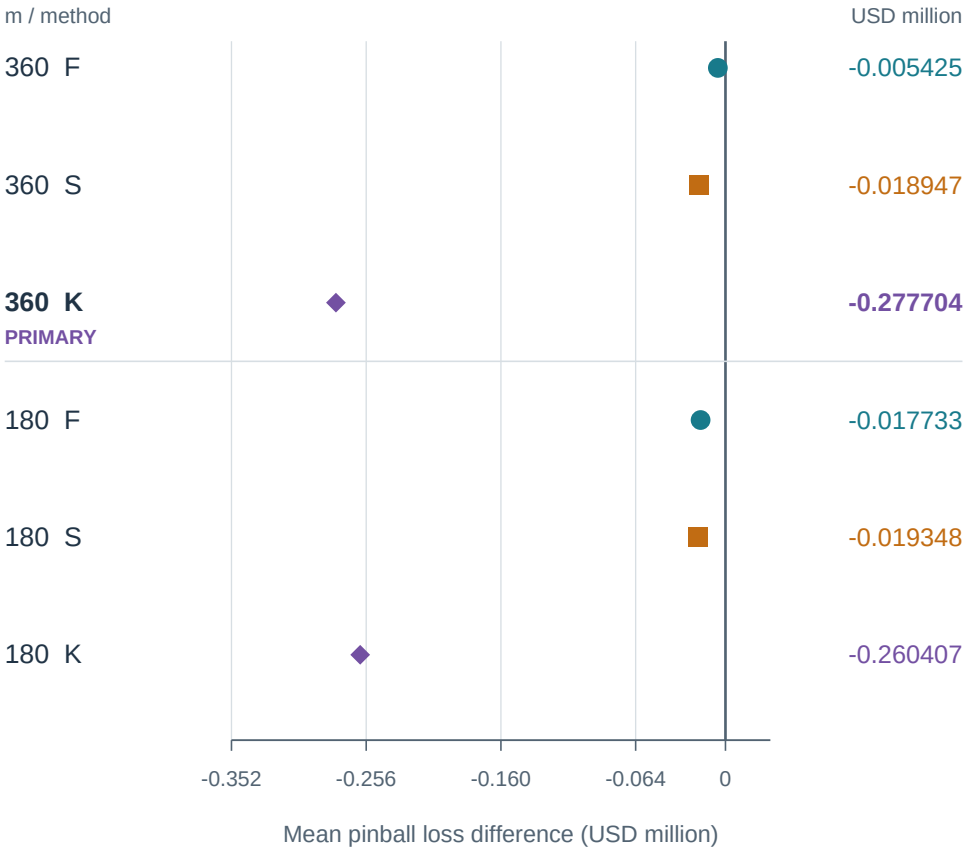
Points are finite-record paired means; the zero line is descriptive, not an inferential threshold.

Nearest versus recent: all frozen loss contrasts

Exploratory paired means | nearest minus recent | negative favors nearest

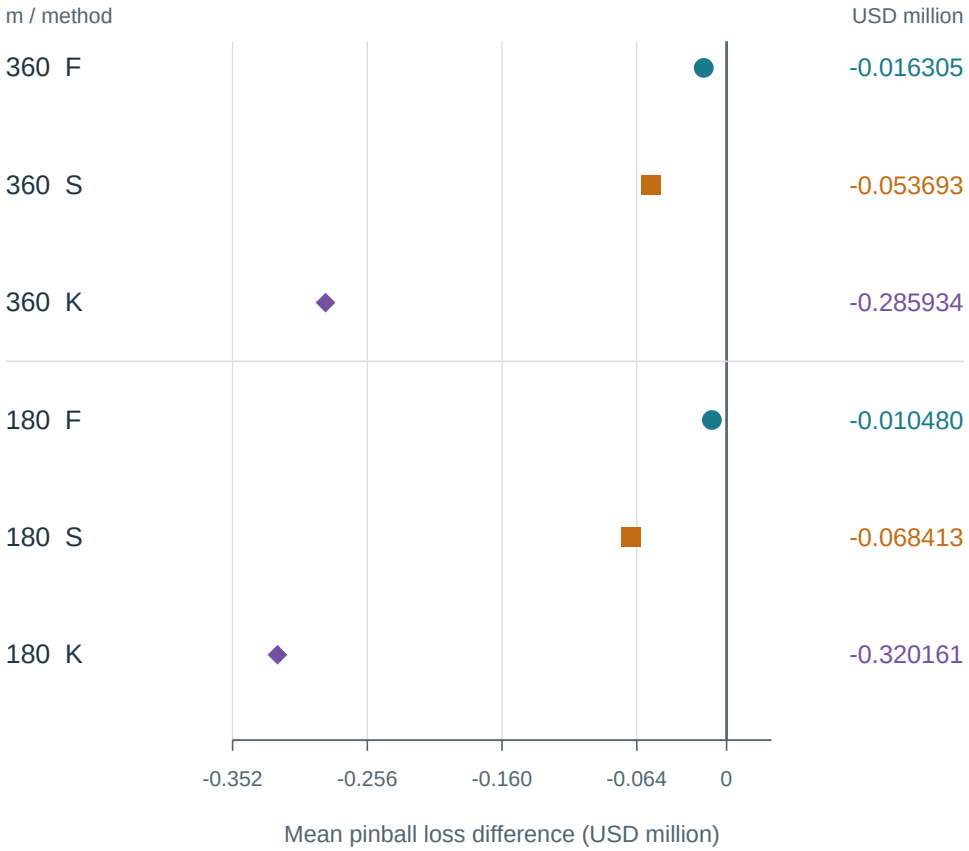
H = 6 records

6,625 paired origins in every row



H = 24 records

6,625 paired origins in every row



F S K

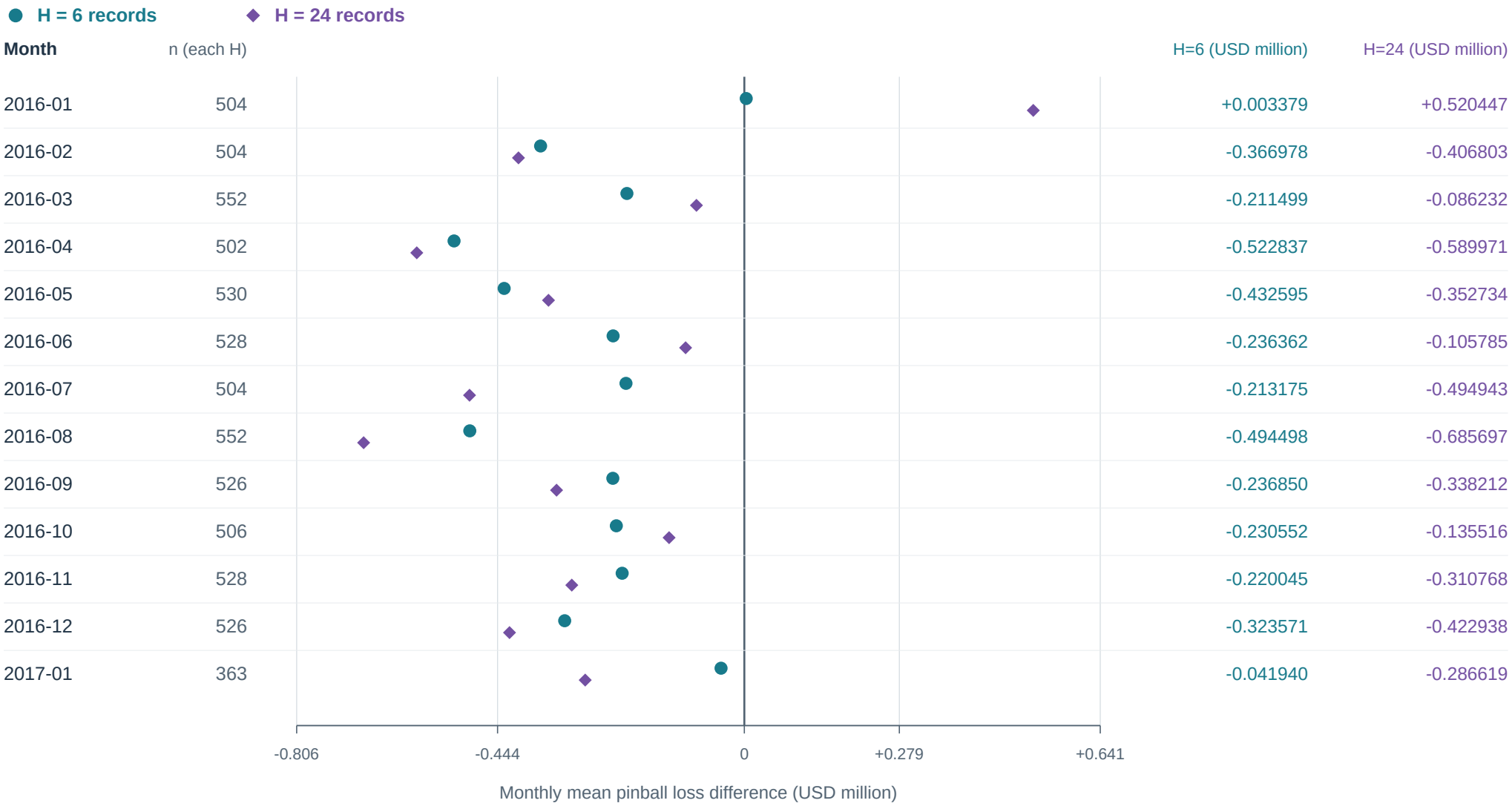
PRIMARY marks H=6, m=360, K, fixed before extension execution; it does not imply an overall ranking.

All 12 contrasts use the already inspected interval. No uncertainty intervals are displayed.

Source: accepted public comparisons.csv | exploratory-state-selection-20261021

Monthly primary contrasts: descriptive / exploratory

Exploratory paired means | nearest360 K minus recent360 K | negative favors nearest



All 13 calendar months are retained, including positive January 2016 at both horizons.

Descriptive summaries of the inspected evaluation record. Zero marks equal average pinball loss.

Source: accepted public monthly.csv | primary nearest360 K minus recent360 K